

数值算法 Homework 05

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Due: 08:00 Mar. 24, 2025

Problem 1

(矩形网格节点下的二维 Lagrange 插值)

Show that 2D Lagrange interpolation over a rectangular mesh is always well-posed, as long as there are no repeated interpolation nodes.

Solution:

首先证明存在性:

当插值节点分布在一个矩形网格上时, 我们可以利用张量积结构构造 2D Lagrange 插值多项式.

设在 x 方向有 $n + 1$ 个互不相同的节点 $x_0, x_1, \dots, x_n$,

在 y 方向上有 $m + 1$ 个互不相同的节点 $y_0, y_1, \dots, y_m$.

总的网格节点为:

$$\{(x_i, y_j) : i = 0, \dots, n, j = 0, \dots, m\}$$

给定二元函数 $f(x, y)$

我们的目标是构造一个多项式 $P(x, y)$ 使得 $P(x_i, y_j) = f(x_i, y_j)$ ($i = 0, \dots, n, j = 0, \dots, m$)

二维 Lagrange 基函数的构造基于一维 Lagrange 基函数:

$$N_i(x) := \prod_{k \neq i}^n \frac{x - x_k}{x_i - x_k} \quad (i = 0, \dots, n)$$

$$M_j(y) := \prod_{k \neq j}^m \frac{y - y_k}{y_j - y_k} \quad (j = 0, \dots, m)$$

$$L_{ij}(x, y) := N_i(x)M_j(y)$$

where $i = 0, \dots, n$ and $j = 0, \dots, m$

容易验证它们满足:

$$L_{ij}(x_{i'}, y_{j'}) = N_i(x_{i'})M_j(y_{j'}) = \begin{cases} 1, & \text{if } i' = i \text{ and } j' = j \\ 0, & \text{otherwise} \end{cases}$$

于是我们可以这样构造二维 Lagrange 插值多项式:

$$\begin{aligned} P(x, y) &:= \sum_{i=0}^n \sum_{j=0}^m f(x_i, y_j) L_{ij}(x, y) \\ &= \sum_{i=0}^n \sum_{j=0}^m f(x_i, y_j) N_i(x) M_j(y) \end{aligned}$$

其次数在 x 方向为 n 次, 在 y 方向为 m 次, 满足:

$$P(x_i, y_j) = f(x_i, y_j)$$

where $i = 0, \dots, n$ and $j = 0, \dots, m$

下面证明唯一性:

假设存在另一个多项式 $Q(x, y)$ 也满足 $Q(x_i, y_j) = f(x_i, y_j)$ ($\forall i, j$),

且其次数在 x 方向不超过 n , 在 y 方向不超过 m

考虑差值多项式 $R(x, y) := P(x, y) - Q(x, y)$, 它在所有节点 (x_i, y_j) 处均为 0.

- 对于每个固定的 y_j , $R(x, y_j)$ 是关于 x 的至多 n 次的一元多项式.
注意到 $R(x, y_j)$ 在 $n + 1$ 个不同点 $x_0, x_1, \dots, x_n$ 处为零.
根据一元多项式唯一性定理, 我们有 $R(x, y_j) \equiv 0$
- 对于每个固定的 x_i , $R(x_i, y)$ 是关于 y 的至多 m 次的一元多项式.
注意到 $R(x_i, y)$ 在 $m + 1$ 个不同点 $y_0, y_1, \dots, y_m$ 处为零.
根据一元多项式唯一性定理, 我们有 $R(x_i, y) \equiv 0$

因此我们有 $R(x, y) \equiv 0$ 成立, 说明多项式 $Q(x, y)$ 与 $P(x, y)$ 完全相同, 唯一性得证.

Problem 2

Let $L[y] := \int_0^1 f(t, y, y') dt$ be a sufficiently smooth functional defined for all sufficiently smooth functions $y(t)$ satisfying $y(0) = y(1) = 0$. Show that any stationary point of $L[y]$ satisfies the **Euler-Lagrange equation**:

$$\frac{\partial f}{\partial y} - \frac{d}{dt} \left(\frac{\partial f}{\partial y'} \right) = 0 \quad (\forall t \in [0, 1])$$

Solution:

设充分光滑的函数 $s(t)$ (满足 $s(0) = s(1) = 0$) 是泛函 $L[y]$ 的驻点.

考虑扰动 $s(t) + \varepsilon \eta(t)$, 其中 $\eta(t)$ 充分光滑, 且满足 $\eta(0) = \eta(1) = 0$, 则我们有:

$$\begin{aligned} \delta L &= \left. \frac{d}{d\varepsilon} L[s + \varepsilon \eta] \right|_{\varepsilon=0} \\ &= \left. \frac{d}{d\varepsilon} \int_0^1 f(t, s + \varepsilon \eta, s' + \varepsilon \eta') dt \right|_{\varepsilon=0} \\ &= \int_0^1 \left(\frac{\partial f}{\partial s} \eta + \frac{\partial f}{\partial s'} \eta' \right) dt \\ &= \int_0^1 \frac{\partial f}{\partial s} \eta dt + \left. \frac{\partial f}{\partial s'} \eta \right|_0^1 - \int_0^1 \frac{d}{dt} \left(\frac{\partial f}{\partial s'} \right) \eta dt \quad (\text{note that } \eta(0) = \eta(1) = 0) \\ &= \int_0^1 \frac{\partial f}{\partial s} \eta dt - \int_0^1 \frac{d}{dt} \left(\frac{\partial f}{\partial s'} \right) \eta dt \\ &= \int_0^1 \left(\frac{\partial f}{\partial s} - \frac{d}{dt} \left(\frac{\partial f}{\partial s'} \right) \right) \eta dt \end{aligned}$$

根据驻点的性质可知 $\delta L = 0$ 对于所有满足 $\eta(0) = \eta(1) = 0$ 的充分光滑的函数 $\eta(t)$ 都成立.

因此我们有 Euler-Lagrange 方程成立:

$$\frac{\partial f}{\partial s} - \frac{d}{dt} \left(\frac{\partial f}{\partial s'} \right) = 0 \quad (\forall t \in [0, 1])$$

Problem 3

Use Gaussian functions (i.e., density of the normal distribution)

to interpolate the Runge function $f(x) = (1 + 25x^2)^{-1}$ over $[-1, 1]$ with a few equispaced nodes.

Visualize the result.

Solution:

给定插值节点 $(x_1, y_1), \dots, (x_n, y_n) \in \mathbb{R} \times \mathbb{R}$ (其中 $x_1, \dots, x_n$ 互不相同)

考虑 Gauss 径向基函数:

$$\begin{aligned}\phi_1(x) &:= \exp(-(x - x_1)^2) \\ \phi_2(x) &:= \exp(-(x - x_2)^2) \\ &\vdots \\ \phi_n(x) &:= \exp(-(x - x_n)^2)\end{aligned}$$

记 $\phi(x) = c_1\phi_1(x) + c_2\phi_2(x) + \cdots + c_n\phi_n(x)$, 则根据插值条件我们有以下线性方程组:

$$\begin{bmatrix} \phi(x_1) \\ \phi(x_2) \\ \vdots \\ \phi(x_n) \end{bmatrix} = \begin{bmatrix} \phi_1(x_1) & \phi_2(x_1) & \cdots & \phi_n(x_1) \\ \phi_1(x_2) & \phi_2(x_2) & \cdots & \phi_n(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(x_n) & \phi_2(x_n) & \cdots & \phi_n(x_n) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

使用 Gauss 径向基函数进行插值的函数:

```
function Gaussian_RBF_interpolation(f, n, interval, gamma)
    x_nodes = linspace(interval(1), interval(2), n);
    y_nodes = f(x_nodes); % 等距采样

    % Construct the interpolation matrix using Gaussian RBFs
    % Gaussian RBF: phi(r) = exp(-(gamma*r).^2)
    A = zeros(n,n);
    for i = 1:n
        for j = 1:n
            r = abs(x_nodes(i) - x_nodes(j));
            A(i,j) = exp(-(gamma * r).^2);
        end
    end

    % Solve for the RBF coefficients
    c = A \ y_nodes';

    % Evaluate the RBF interpolation at each point in x_exact
    x_exact = linspace(interval(1), interval(2), 1000);
    y_exact = f(x_exact); % 精确值
    y_interp = zeros(size(x_exact)); % 插值结果
    for i = 1:length(x_nodes)
        r_vec = abs(x_exact - x_nodes(i));
        phi = exp(-(gamma * r_vec).^2);
        y_interp = y_interp + c(i) * phi;
    end

    % 插值与精确值的差距
    y_diff = abs(y_interp - y_exact);

    % 绘图
    % 左侧纵轴: 绘制精确值和插值结果
    yyaxis left;
    plot(x_exact, y_exact, 'k-', 'Linewidth', 1.5); hold on;
    plot(x_exact, y_interp, 'r--', 'Linewidth', 1);
    scatter(x_nodes, y_nodes, 25, 'k', 'filled');
    title(['Gaussian RBF Interpolation with n = ', num2str(n)]);
    hold off;

    % 右侧纵轴: 绘制 log(y_diff)
    yyaxis right;
```

```

plot(x_exact, log(y_diff), 'b-.', 'Linewidth', 1);
ylabel('log(y_{diff})');
legend('Exact', 'Interpolated', 'Nodes', 'log difference');
grid on;
end

```

函数调用:

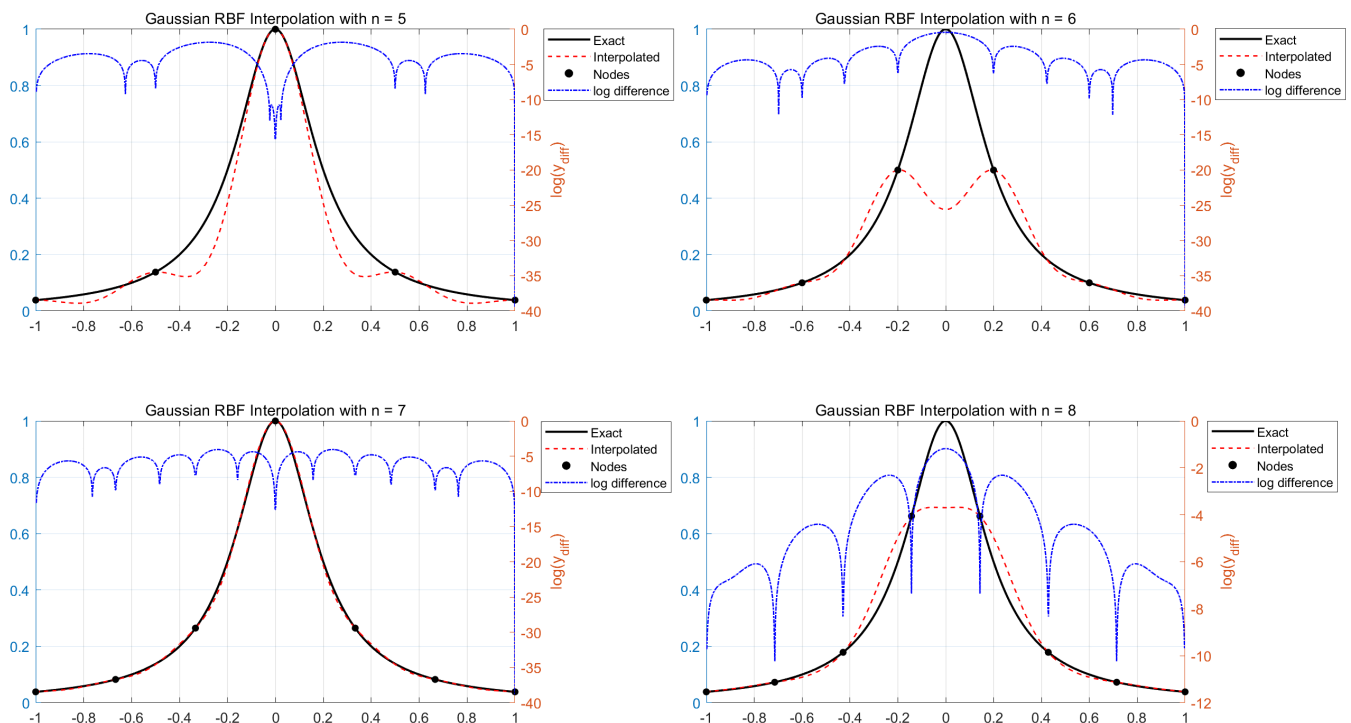
```

f = @(x) 1 ./ (1 + 25 * x.^2);
ns_f = [5, 6, 7, 8];
gamma = 5; % Gauss 径向基函数的参数

figure;
for i = 1:length(ns_f)
    subplot(2,2,i);
    Gaussian_RBF_interpolation(f, ns_f(i), [-1, 1], gamma);
end

```

运行结果:



显然在使用 Gauss 径向基函数进行插值的时候, 为被插函数的每个峰取一个节点是必要的.
而且 Gauss 径向基函数比较适合刻画对称的峰.

Problem 4

Interpolate the following data set and visualize your solution on $[-1, 1] \times [-1, 1]$.

x_i	y_i	z_i
-1.0000	-1.0000	1.6389
-1.0000	1.0000	0.5403
1.0000	-1.0000	-0.9900
1.0000	1.0000	0.1086
-0.7313	0.6949	0.9573
0.5275	-0.4899	0.8270
-0.0091	-0.1010	1.6936
0.3031	0.5774	1.3670

Note that different interpolation strategies will lead to different results.

You are encouraged to try different 2D interpolation strategies learned from the lecture.

(You can use MATLAB/Octave function `de1aunay` to perform Delaunay triangularization if needed)

(1) Shepard 插值

给定插值节点 $x_1, \dots, x_n \in \mathbb{R}^d$, 其 Shepard 插值多项式为:

$$\phi(x) := \sum_{i=1}^n f(x_i) \phi_i(x)$$

$$\text{where } \phi_i(x) := \frac{\|x - x_i\|^{-p}}{\sum_{k=1}^n \|x - x_k\|^{-p}} \quad (i = 1, \dots, n)$$

其中 $\|\cdot\|$ 可以是 $\mathbb{R}^d$ 上的任意范数, 常数 $p \geq 1$

在编程时, 我们直接进行以下赋值, 以避免 ∞/∞ 的现象:

$$\phi_i(x_k) = \begin{cases} 1, & \text{if } k = i \\ 0, & \text{otherwise} \end{cases}$$

Matlab 代码:

```
% Data: Each row is [x, y, f(x,y)]
nodes = [-1.0000, -1.0000, 1.6389;
         -1.0000,  1.0000, 0.5403;
          1.0000, -1.0000, -0.9900;
          1.0000,  1.0000, 0.1086;
         -0.7313,  0.6949, 0.9573;
          0.5275, -0.4899, 0.8270;
         -0.0091, -0.1010, 1.6936;
          0.3031,  0.5774, 1.3670];

% Extract x and y coordinates of nodes and their function values
x_nodes = nodes(:,1);
y_nodes = nodes(:,2);
f_nodes = nodes(:,3);

% Parameters for Shepard interpolation
p = 2;           % power parameter (p>=1)
tol = 1e-12;     % tolerance to determine if a grid point is exactly at a node

% Define a grid over [-1,1] x [-1,1]
numPts = 300; % Number of grid points in each direction
[x_grid, y_grid] = meshgrid(linspace(-1, 1, numPts), linspace(-1, 1, numPts));
gridPts = [x_grid(:), y_grid(:)]; % Each row is a point in the grid
```

```

% Initialize the interpolated values at grid points
f_interp = zeros(size(gridPts,1), 1);

% Shepard interpolation over the grid
% For each grid point, compute distances to all nodes, then apply the formula:
%   phi(x) = sum_i [ f(x_i)*w_i(x) ] / sum_i [w_i(x) ]
% where w_i(x) = 1/||x - x_i||^p.
%
% Also, if the grid point is exactly at a node (distance < tol),
% then assign the corresponding node value.
for i = 1:size(gridPts,1)
    % Current grid point (x,y)
    pt = gridPts(i, :);

    % Compute Euclidean distances from pt to each node
    distances = sqrt((pt(1) - x_nodes).^2 + (pt(2) - y_nodes).^2);

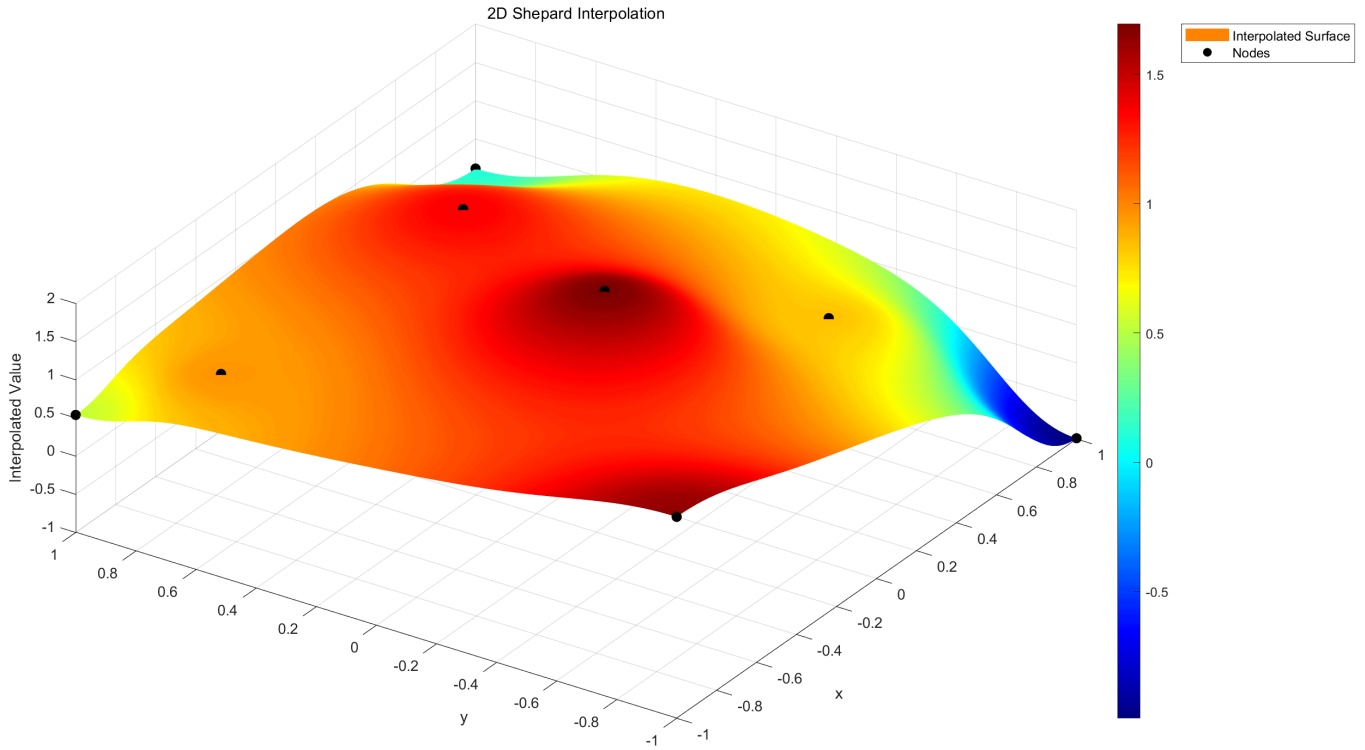
    % Check if pt coincides with any node (within tolerance)
    idx = find(distances < tol, 1);
    if ~isempty(idx)
        % If the grid point equals a node, assign its function value directly
        f_interp(i) = f_nodes(idx);
    else
        % Compute weights: w_i = 1 / (distance^p)
        w = 1 ./ (distances.^p);
        % Normalize the weights so that they sum to 1
        lambda = w / sum(w);
        % Shepard interpolation: weighted sum of node function values
        f_interp(i) = sum(lambda .* f_nodes);
    end
end

% Reshape the interpolation result to the grid shape
F_interp = reshape(f_interp, size(x_grid));

% Visualization of the interpolated surface
figure;
surf(x_grid, y_grid, F_interp, 'EdgeColor', 'none');
hold on;
scatter3(x_nodes, y_nodes, f_nodes, 50, 'k', 'filled');
title('2D Shepard Interpolation');
xlabel('x');
ylabel('y');
zlabel('Interpolated Value');
colormap jet;
colorbar;
legend('Interpolated Surface', 'Nodes');
grid on;
view(3);
hold off;

```

运行结果:



(2) 重心坐标插值

三角网格节点下的二维 Bernstein 插值:

在三角形区域内进行多项式插值通常不采用张量积结构, 而是借助**重心坐标** (barycentric coordinates) 进行构造.

考虑三角形 $\triangle ABC$

设其顶点坐标为 $A = (x_A, y_A), B = (x_B, y_B), C = (x_C, y_C)$

三角形 $\triangle ABC$ 上的任意一点 $Q = (x, y)$ 都可用重心坐标表示:

$$Q = \alpha A + \beta B + \gamma C$$

$$\Leftrightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x_A & x_B & x_C \\ y_A & y_B & y_C \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}$$

$$\text{where } \begin{cases} \alpha, \beta, \gamma \geq 0 \\ \alpha + \beta + \gamma = 1 \end{cases}$$

记 A, B, C 的重心坐标为 $(1, 0, 0), (0, 1, 0), (0, 0, 1)$

给定多项式次数 d

若我们在三角形 $\triangle ABC$ 按重心坐标等距取点:

$$Q_{i,j,k} = \left(\frac{i}{d}, \frac{j}{d}, \frac{k}{d} \right)$$

$$\text{where } \begin{cases} 0 \leq i, j, k \leq d \\ i + j + k = d \end{cases}$$

$$f(Q_{i,j,k}) = f\left(\frac{i}{d}x_A + \frac{j}{d}x_B + \frac{k}{d}x_C, \frac{i}{d}y_A + \frac{j}{d}y_B + \frac{k}{d}y_C\right)$$

则 $Q = (\alpha, \beta, \gamma)$ 的 d 次多项式插值为:

$$\begin{aligned}
P(Q) &= P(\alpha, \beta, \gamma) \\
&= \sum_{\substack{0 \leq i, j, k \leq d \\ i+j+k=d}} f(Q_{i,j,k}) B_{i,j,k}(Q) \\
&= \sum_{\substack{0 \leq i, j, k \leq d \\ i+j+k=d}} f(Q_{i,j,k}) B_{i,j,k}(\alpha, \beta, \gamma)
\end{aligned}$$

其中 Bernstein 多项式的定义如下:

$$\begin{aligned}
B_{i,j,k}(\alpha, \beta, \gamma) &= \frac{d!}{i!j!k!} \alpha^i \beta^j \gamma^k \\
\text{where } &\begin{cases} 0 \leq i, j, k \leq d \\ i + j + k = d \end{cases}
\end{aligned}$$

特殊地, $d = 1$ 时就是线性插值:

$$\begin{aligned}
B_{1,0,0}(\alpha, \beta, \gamma) &= \alpha \\
B_{0,1,0}(\alpha, \beta, \gamma) &= \beta \\
B_{0,0,1}(\alpha, \beta, \gamma) &= \gamma
\end{aligned}$$

$$\begin{aligned}
P(Q) &= P(\alpha, \beta, \gamma) \\
&= \alpha f(A) + \beta f(B) + \gamma f(C) \\
&= \alpha f(x_A, y_A) + \beta f(x_B, y_B) + \gamma f(x_C, y_C)
\end{aligned}$$

Matlab 代码:

```

% Given Data: Each row is [x, y, f(x,y)]
nodes = [-1.0000, -1.0000, 1.6389;
         -1.0000,  1.0000, 0.5403;
          1.0000, -1.0000, -0.9900;
          1.0000,  1.0000, 0.1086;
         -0.7313,  0.6949, 0.9573;
          0.5275, -0.4899, 0.8270;
         -0.0091, -0.1010, 1.6936;
          0.3031,  0.5774, 1.3670];

% Extract x, y and function values
x_nodes = nodes(:,1);
y_nodes = nodes(:,2);
f_nodes = nodes(:,3);

% Delaunay Triangulation and create triangulation object
tri = delaunay(x_nodes, y_nodes);
tr = triangulation(tri, x_nodes, y_nodes); % 创建triangulation对象

% 绘图部分保持不变
figure;
triplot(tr, 'k-'); hold on;
scatter(x_nodes, y_nodes, 80, f_nodes, 'filled');
colorbar;
title('Delaunay Triangulation of Nodes');
xlabel('x');
ylabel('y');
axis equal;
hold off;

% 定义插值网格

```



```

numPts = 300;
[x_grid, y_grid] = meshgrid(linspace(-1,1,numPts), linspace(-1,1,numPts));
gridPoints = [x_grid(:), y_grid(:)];

% 使用pointLocation查找三角形索引
tri_idx = pointLocation(tr, gridPoints);

% 初始化插值结果
f_interp = nan(size(gridPoints,1), 1);

% 遍历每个网格点进行插值
for i = 1:length(tri_idx)
    ti = tri_idx(i);
    if ~isnan(ti)
        % 获取三角形顶点索引
        vertexInd = tri(ti, :);
        fv = f_nodes(vertexInd);

        % 使用内置函数计算重心坐标
        bary = cartesianToBarycentric(tr, ti, gridPoints(i, :));

        % 确保数值稳定性（处理舍入误差）
        bary = max(bary, 0); % 避免负值
        bary = bary / sum(bary); % 确保归一化

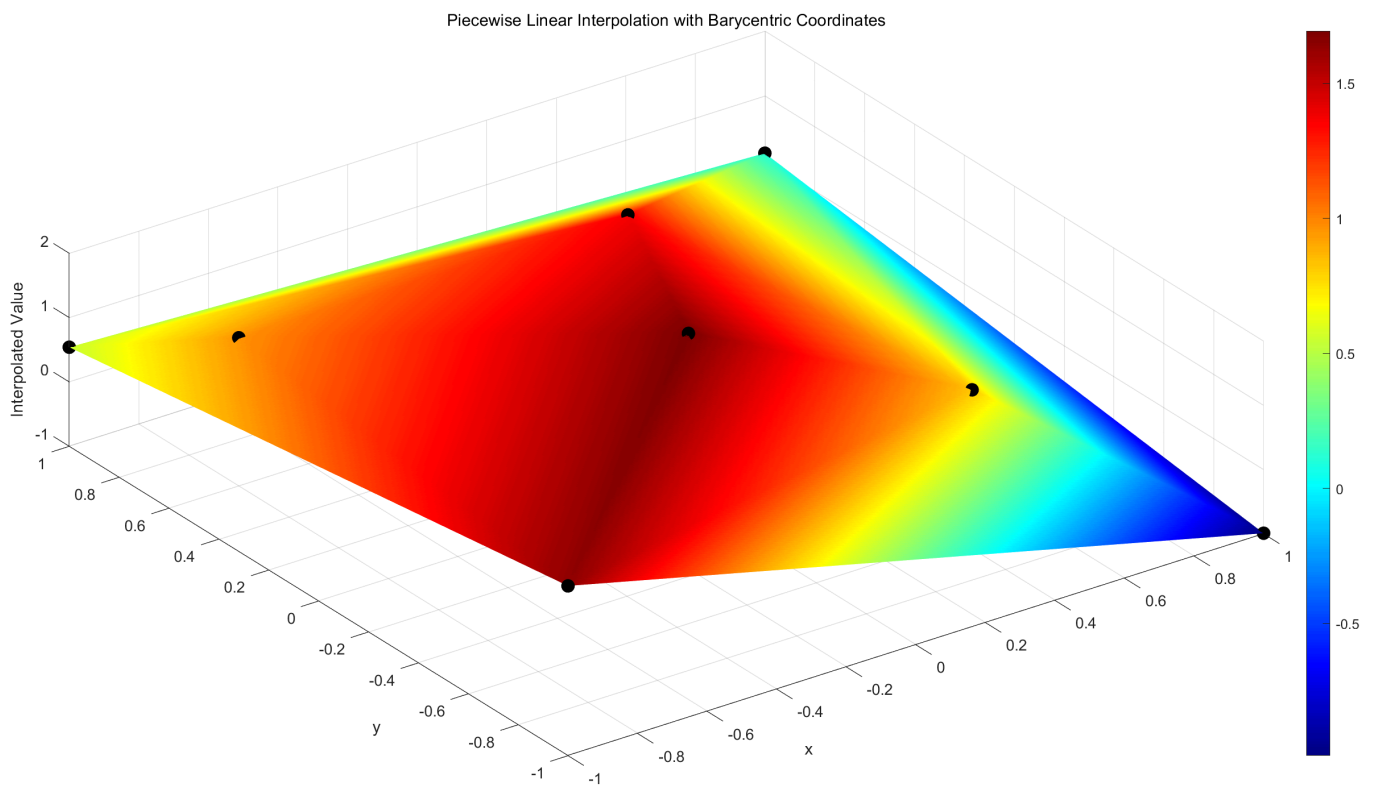
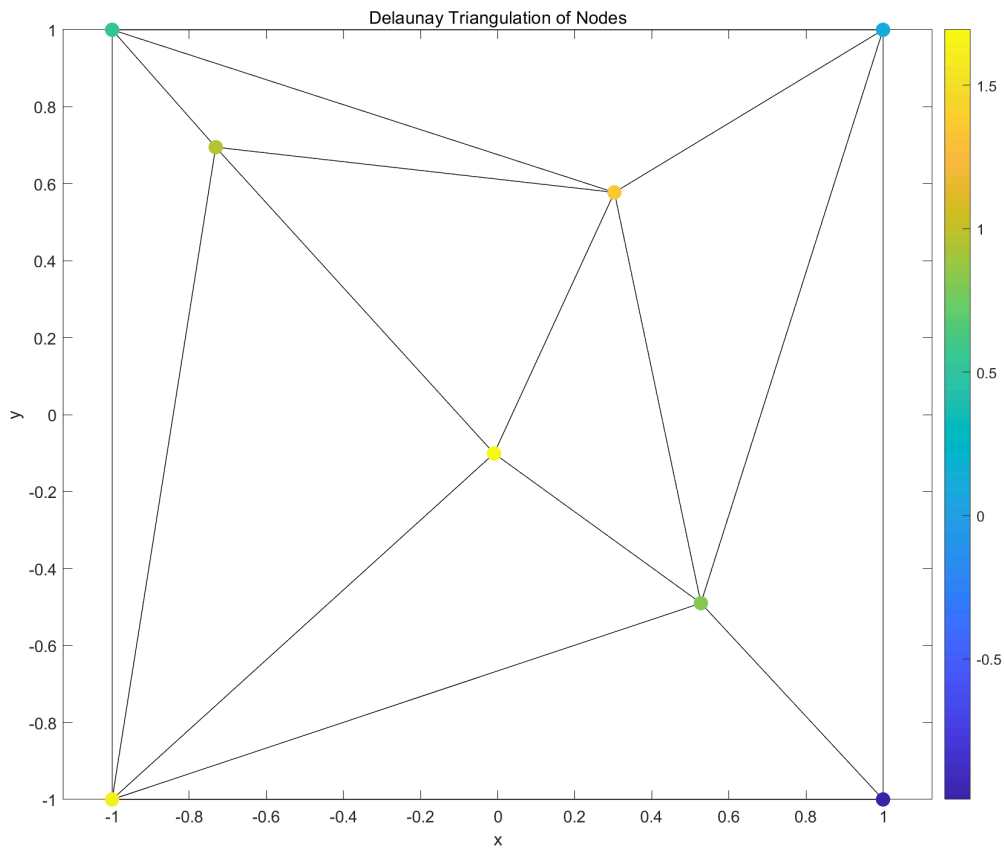
        % 线性插值
        f_interp(i) = bary * fv;
    end
end

% 重构网格数据并绘图
F_interp = reshape(f_interp, size(x_grid));

figure;
surf(x_grid, y_grid, F_interp, 'EdgeColor', 'none');
hold on;
scatter3(x_nodes, y_nodes, f_nodes, 80, 'k', 'filled');
title('Piecewise Linear Interpolation with Barycentric Coordinates');
xlabel('x'); ylabel('y'); zlabel('Interpolated Value');
colormap jet; colorbar;
view(3);
grid on;
hold off;

```

运行结果:



Problem 5 (optional)

In the lecture we have proved the minimum energy property of cubic splines.
Can you derive the minimum energy property for certain types of quintic splines?

Solution:

设样条 $s(x)$ 在区间 $[a, b]$ 上, 型值点 (x_i, y_i) ($i = 0, \dots, n$) 的横坐标为 $a = x_0 < x_1 < \dots < x_n = b$
 其中 $y_i = f(x_i)$ ($i = 0, \dots, n$)

假设 $s(x)$ 在相邻型值点之间为五次多项式, 即 $s(x)$ 在区间 $[x_i, x_{i+1}]$ 上的表达式为:

$$s_i(x) := a_i + b_i x + c_i x^2 + d_i x^3 + e_i x^4 + f_i x^5 \quad (i = 0, \dots, n-1)$$

因此 $s(x)$ 在整个 $[a, b]$ 区间中有 $6n$ 个待定系数.

它至少要满足以下条件:

- ① 通过型值点: (共 $2n$ 个方程)

$$\begin{cases} s_i(x_i) = y_i \\ s_i(x_{i+1}) = y_{i+1} \end{cases} \quad (i = 0, \dots, n-1)$$

- ② 在型值点处的一、二、三、四阶导数连续: (共 $4(n-1)$ 个方程)

$$s_i^{(k)}(x_{i+1}) = s_{i+1}^{(k)}(x_{i+1}) \quad (i = 0, \dots, n-2)$$

where $k = 1, 2, 3, 4$

上面只列出了 $6n - 4$ 个方程, 而总共有 $6n$ 个待定系数.

为使 $s(x)$ 能够被唯一确定, 我们通常会补充边界条件.

以下是三种典型的边界条件:

- ① **完全样条 (Complete Spline):**

假设已知样条在边界点 x_0, x_n 处的切线斜率值分别为 k_0, k_n , 曲率值为 m_0, m_n :

$$\begin{aligned} s'_0(x_0) &= k_0, \quad s''_0(x_0) = m_0 \\ s'_{n-1}(x_n) &= k_n, \quad s''_{n-1}(x_n) = m_n \end{aligned}$$

where $k_0 = f'(x_0), k_n = f'(x_n), m_0 = f''(x_0), m_n = f''(x_n)$

- ② **自然样条 (Natural Spline):**

假设样条在边界点 x_0, x_n 处的三、四阶导数为零:

$$\begin{aligned} s'''_0(x_0) &= s'''_{n-1}(x_n) = 0 \\ s^{(4)}_0(x_0) &= s^{(4)}_{n-1}(x_n) = 0 \end{aligned}$$

- ③ **周期样条 (Periodic Spline):**

假设样条的周期为 $b - a = x_n - x_0$, 要求边界点 x_0 和 x_n 处的衔接是光滑的:

$$\begin{aligned} s'_0(x_0) &= s'_{n-1}(x_n) \\ s''_0(x_0) &= s''_{n-1}(x_n) \\ s'''_0(x_0) &= s'''_{n-1}(x_n) \\ s^{(4)}_0(x_0) &= s^{(4)}_{n-1}(x_n) \end{aligned}$$

- ④ **Not-a-knot 样条:**

消除样条在 $x_1, x_2, x_{n-2}, x_{n-1}$ 处的 "结", 即要求五阶导数在 $x_1, x_2, x_{n-2}, x_{n-1}$ 处连续.

换言之, 我们要求 $[x_0, x_1], [x_1, x_2], [x_2, x_3]$ 上的五次多项式 $s_0(x), s_1(x), s_2(x)$ 相同,

同时要求 $[x_{n-3}, x_{n-2}], [x_{n-2}, x_{n-1}], [x_{n-1}, x_n]$ 上的五次多项式 $s_{n-3}(x), s_{n-2}(x), s_{n-1}(x)$ 相同.

$$\begin{aligned} s^{(5)}_0(x_1) &= s^{(5)}_1(x_1) \\ s^{(5)}_1(x_2) &= s^{(5)}_2(x_2) \\ s^{(5)}_{n-3}(x_{n-2}) &= s^{(5)}_{n-2}(x_{n-2}) \\ s^{(5)}_{n-2}(x_{n-1}) &= s^{(5)}_{n-1}(x_{n-1}) \end{aligned}$$

五次样条能够最小化势能 $\|s^{(3)}(x)\|_2^2 := \int_a^b (s^{(3)}(x))^2 dx$

具体来说, 对于任意给定的光滑函数 $f: \mathbb{R} \mapsto \mathbb{R}$, 定义以下线性空间:

$$\begin{aligned}\mathcal{W}_0 &:= \{g \in C^3([a, b]) : g(x_i) = f(x_i) \text{ for } i = 0, 1, \dots, n\} \\ \mathcal{W}_1 &:= \left\{g \in \mathcal{W}_0 : \begin{cases} g'(x_0) = f'(x_0), g'(x_n) = f'(x_n) \\ g''(x_0) = f''(x_0), g''(x_n) = f''(x_n) \end{cases}\right\} \\ &\text{where } a = x_0 < x_1 < \dots < x_n = b\end{aligned}$$

可以证明:

- ① 若 $s(x)$ 是 $f(x)$ 的五次自然样条 (注意 $s(x)$ 也属于 $\mathcal{W}_0$), 则对于任意 $g \in \mathcal{W}_0$ 都有 $\|s^{(3)}(x)\|_2 \leq \|g^{(3)}(x)\|_2$ (这表明在没有额外边界约束的情况下, 五次自然样条的势能最小)
- ② 若 $s(x)$ 是 $f(x)$ 的五次完全样条 (注意 $s(x)$ 也属于 $\mathcal{W}_1$), 则对于任意 $g \in \mathcal{W}_1$ 都有 $\|s^{(3)}(x)\|_2 \leq \|g^{(3)}(x)\|_2$ (这表明在有额外的边界点一、二阶导数约束的情况下, 五次完全样条的势能最小)

证明思路:

记泛函 $F[g] := \|g^{(3)}\|_2^2 = \int_a^b (g^{(3)}(x))^2 dx$

我们考虑将多元最值问题转化为一元最值问题.

任意给定 $u(x)$, 我们都有:

$$\begin{aligned}F[s + tu] &= \|s^{(3)} + tu^{(3)}\|_2^2 \\ &= \|s^{(3)}\|_2^2 + 2t\langle s^{(3)}, u^{(3)} \rangle_2 + t^2\|u^{(3)}\|_2^2 \\ &= F[s] + 2t \int_a^b s^{(3)}(x)u^{(3)}(x)dx + t^2F[u]\end{aligned}$$

对于第一个命题:

要证明 $F[s] \leq F[g]$ 对任意 $g \in \mathcal{W}_0$ 都成立,

即要证明 $F[s + tu] \geq F[s]$ 对于任意 $u \in \mathcal{W}_0$ (注意 $\mathcal{W}_0$ 是线性空间, 自然样条 s 也属于 $\mathcal{W}_0$) 和 $t \in \mathbb{R}$ 都成立,

即要证明 $\int_a^b s^{(3)}(x)u^{(3)}(x)dx = 0$ 对于任意 $u \in \mathcal{W}_0$ 都成立.

第二个命题的证明也是类似的.

任意给定三阶可导函数 $u \in C^3([a, b])$

我们要对积分 $\int_a^b s^{(3)}(x)u^{(3)}(x)dx$ 进行分段处理,

这是因为五次样条函数 $s(x)$ 是 "分段光滑的",

具体来说, 它的五阶导数和六阶导数在插值点 $x_1, \dots, x_{n-1}$ 处不一定存在.

每段积分 $\int_{x_i}^{x_{i+1}} s^{(3)}(x)u^{(3)}(x)dx$ 可以使用分部积分公式进行变形处理:

$$\begin{aligned}\int_a^b s^{(3)}(x)u^{(3)}(x)dx &= \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} s^{(3)}(x)u^{(3)}(x)dx \\ &= \sum_{i=0}^{n-1} \left[s^{(3)}(x)u^{(2)}(x) \Big|_{x_i}^{x_{i+1}} - \int_{x_i}^{x_{i+1}} s^{(4)}(x)u^{(2)}(x)dx \right] \\ &= \sum_{i=0}^{n-1} \left[(s^{(3)}(x)u^{(2)}(x) - s^{(4)}(x)u^{(1)}(x)) \Big|_{x_i}^{x_{i+1}} + \int_{x_i}^{x_{i+1}} s^{(5)}(x)u^{(1)}(x)dx \right] \\ &= \sum_{i=0}^{n-1} \left[(s^{(3)}(x)u^{(2)}(x) - s^{(4)}(x)u^{(1)}(x) + s^{(5)}(x)u(x)) \Big|_{x_i}^{x_{i+1}} + \int_{x_i}^{x_{i+1}} s^{(6)}(x)u(x)dx \right] \\ &\quad (\text{note that } s^{(6)}(x) = 0 \text{ for } x \in (x_i, x_{i+1}) \text{ where } i = 0, 1, \dots, n-1) \\ &= [s^{(3)}(b)u^{(2)}(b) - s^{(4)}(b)u^{(1)}(b) + s^{(5)}(b)u(b)] - [s^{(3)}(a)u^{(2)}(a) - s^{(4)}(a)u^{(1)}(a) + s^{(5)}(a)u(a)] \\ &= [s^{(3)}(b)u^{(2)}(b) - s^{(3)}(a)u^{(2)}(a)] - [s^{(4)}(b)u^{(1)}(b) - s^{(4)}(a)u^{(1)}(a)] + [s^{(5)}(b)u(b) - s^{(5)}(a)u(a)]\end{aligned}$$

- **对于第一个命题:**

设 $s(x)$ 是五次自然样条, 则其边界条件为 $\begin{cases} s^{(3)}(a) = s^{(3)}(b) = 0 \\ s^{(4)}(a) = s^{(4)}(b) = 0 \end{cases}$

任意给定 $g \in \mathcal{W}_0$, 取 $u(x) = g(x) - s(x)$, 则根据插值条件有 $u(a) = u(b) = 0$
于是我们有:

$$\begin{aligned}\int_a^b s^{(3)}(x)u^{(3)}(x)dx &= [s^{(3)}(b)u^{(2)}(b) - s^{(3)}(a)u^{(2)}(a)] - [s^{(4)}(b)u^{(1)}(b) - s^{(4)}(a)u^{(1)}(a)] + [s^{(5)}(b)u(b) - s^{(5)}(a)u(a)] \\ &= [0 \cdot u^{(2)}(b) - 0 \cdot u^{(2)}(a)] - [0 \cdot u^{(1)}(b) - 0 \cdot u^{(1)}(a)] + [s^{(5)}(b) \cdot 0 - s^{(5)}(a) \cdot 0] \\ &= 0\end{aligned}$$

根据 "证明思路" 中的推理可知对于任意 $g \in \mathcal{W}_0$ 我们都有 $\|s^{(3)}\|_2 \leq \|g^{(3)}\|_2$ 成立.

• **对于第二个命题:**

设 $s(x)$ 是五次完全样条, 则其边界条件为 $\begin{cases} s'(a) = f'(a), s'(b) = f'(b) \\ s''(a) = f''(a), s''(b) = f''(b) \end{cases}$

任意给定 $g \in \mathcal{W}_1$, 取 $u(x) = g(x) - s(x)$, 则根据插值条件有 $u(a) = u(b) = 0$

同时根据 $\mathcal{W}_1$ 的定义有 $\begin{cases} u'(a) = u'(b) = 0 \\ u''(a) = u''(b) = 0 \end{cases}$

于是我们有:

$$\begin{aligned}\int_a^b s^{(3)}(x)u^{(3)}(x)dx &= [s^{(3)}(b)u^{(2)}(b) - s^{(3)}(a)u^{(2)}(a)] - [s^{(4)}(b)u^{(1)}(b) - s^{(4)}(a)u^{(1)}(a)] + [s^{(5)}(b)u(b) - s^{(5)}(a)u(a)] \\ &= [s^{(3)}(b) \cdot 0 - s^{(3)}(a) \cdot 0] - [s^{(4)}(b) \cdot 0 - s^{(4)}(a) \cdot 0] + [s^{(5)}(b) \cdot 0 - s^{(5)}(a) \cdot 0] \\ &= 0\end{aligned}$$

根据 "证明思路" 中的推理可知对于任意 $g \in \mathcal{W}_1$ 我们都有 $\|s^{(3)}\|_2 \leq \|g^{(3)}\|_2$ 成立.

Problem 6 (optional)

Let us consider Problem 4 from last week again.

Can you give a theoretical analysis on how much error Joe will make on each subinterval?

(Homework 04 Problem 04)

When performing interpolation with a complete cubic spline, the choice of derivatives on the boundary is important.

Suppose that Joe wants to interpolate the sine function $f(x) = \sin(x)$ at nine equispaced nodes over $[0, 2\pi]$, with $f'(0) = f'(2\pi) = 1$.

Unfortunately, he made a typo on $f'(0)$ in his program and observed some strange results.

Try to reproduce Joe's result with a few different values of $f'(0)$.

For instance, $f'(0) = 0, f'(0) = -1$, etc.

Solution:

三次完全样条要求解的是这样一个稀疏线性方程组:

$$\begin{bmatrix} \alpha_1 & \beta_1 & & & \\ \beta_1 & \alpha_2 & \beta_2 & & \\ & \beta_2 & \ddots & \ddots & \\ & & \ddots & \alpha_{n-2} & \beta_{n-2} \\ & & & \beta_{n-2} & \alpha_{n-1} \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_{n-2} \\ k_{n-1} \end{bmatrix} = \begin{bmatrix} \eta_0 + \eta_1 - \beta_0 k_0 \\ \eta_1 + \eta_2 \\ \vdots \\ \eta_{n-3} + \eta_{n-2} \\ \eta_{n-2} + \eta_{n-1} - \beta_{n-1} k_n \end{bmatrix}$$

$$\text{where } \begin{cases} h_i = x_{i+1} - x_i & (i = 0, \dots, n-1) \\ \beta_i = \frac{1}{h_i} & (i = 0, \dots, n-1) \\ \alpha_{i+1} = 2(\beta_i + \beta_{i+1}) & (i = 0, \dots, n-2) \\ \eta_i = 3 \frac{y_{i+1} - y_i}{h_i^2} & (i = 0, \dots, n-1) \end{cases}$$

其中 $x_0, x_1, \dots, x_n$ 是插值节点.

在等距节点的假设下，部分变量的形式可以得到简化：

$$\begin{aligned}x_{i+1} - x_i &\equiv h \\ \beta &= \frac{1}{h} \\ \alpha &= 2(\beta + \beta) = 4\beta\end{aligned}$$

于是上述稀疏线性方程组变为：

$$\begin{bmatrix} 4\beta & \beta & & & \\ \beta & 4\beta & \beta & & \\ & \beta & \ddots & \ddots & \\ & & \ddots & 4\beta & \beta \\ & & & \beta & 4\beta \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_{n-2} \\ k_{n-1} \end{bmatrix} = \begin{bmatrix} \eta_0 + \eta_1 - \beta k_0 \\ \eta_1 + \eta_2 \\ \vdots \\ \eta_{n-3} + \eta_{n-2} \\ \eta_{n-2} + \eta_{n-1} - \beta k_n \end{bmatrix}$$

where $\beta = \frac{1}{h}$ and $\eta_i = 3 \frac{y_{i+1} - y_i}{h^2}$ ($i = 0, \dots, n-1$)

正如我们在 **Homework 04 Problem 04** 中看到的那样，

三次样条插值的边界条件错误带来的影响一定是全局的，但到远端会逐步衰减。

这是因为此问题的系数矩阵 βA 的逆矩阵 $\frac{1}{\beta} A^{-1}$ 的元素越靠近主对角线就越大。

误差是逆矩阵 $\frac{1}{\beta} A^{-1}$ 乘右端项 b ，于是右端项第一个元素的偏差对解的第一个元素影响大，对解的后续元素影响小。

$$\text{Denote } A = \begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & 1 & \ddots & \ddots & \\ & & \ddots & 4 & 1 \\ & & & 1 & 4 \end{bmatrix}_{(n-1) \times (n-1)}$$

(1) 行列式

首先考虑对称三对角阵的行列式递推关系：

$$\begin{aligned}
& \text{Define } D_0 = 1 \\
& D_1 = |\alpha_1| = \alpha_1 \\
D_n &= \begin{vmatrix} \alpha_1 & \beta_1 & & & \\ \beta_1 & \alpha_2 & & & \\ & \ddots & \ddots & & \\ & & \beta_{n-3} & \alpha_{n-2} & \beta_{n-2} \\ & & & \beta_{n-2} & \alpha_{n-1} & \beta_{n-1} \\ & & & & \beta_{n-1} & \alpha_n \end{vmatrix} \quad (\text{decompose it using the last line}) \\
&= \alpha_n \begin{vmatrix} \alpha_1 & \beta_1 & & & \\ \beta_1 & \alpha_2 & & & \\ & \ddots & \ddots & & \\ & & \beta_{n-3} & \alpha_{n-2} & \beta_{n-2} \\ & & & \beta_{n-2} & \alpha_{n-1} \end{vmatrix} - \beta_{n-1} \begin{vmatrix} \alpha_1 & \beta_1 & & & \\ \beta_1 & \alpha_2 & & & \\ & \ddots & \ddots & & \\ & & \beta_{n-3} & \alpha_{n-2} & 0 \\ & & & \beta_{n-2} & \beta_{n-1} \end{vmatrix} \\
&= \alpha_n D_{n-1} - \beta_{n-1} \cdot \beta_{n-1} \begin{vmatrix} \alpha_1 & \beta_1 & & & \\ \beta_1 & \alpha_2 & & & \\ & \ddots & \ddots & & \\ & & \beta_{n-3} & \alpha_{n-2} \end{vmatrix} \\
&= \alpha_n D_{n-1} - \beta_{n-1}^2 D_{n-2}
\end{aligned}$$

现取:

$$A = \begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & & & \\ & 1 & \ddots & & \\ & & \ddots & 4 & 1 \\ & & & 1 & 4 \end{bmatrix}_{n \times n}$$

此时行列式的递推关系为:

$$\begin{aligned}
D_{-1} &= 0 \\
D_0 &= 1 \\
D_1 &= 4 \\
D_n &= 4D_{n-1} - D_{n-2} \\
&\Downarrow \\
\begin{bmatrix} D_n \\ D_{n-1} \end{bmatrix} &= \begin{bmatrix} 4 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} D_{n-1} \\ D_{n-2} \end{bmatrix} \text{ where } \begin{bmatrix} D_0 \\ D_{-1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}
\end{aligned}$$

我们对系数矩阵 $\begin{bmatrix} 4 & -1 \\ 1 & 0 \end{bmatrix}$ 进行谱分解:

$$\begin{aligned}
\begin{bmatrix} 4 & -1 \\ 1 & 0 \end{bmatrix} &= \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \phi_1 & \\ & \phi_2 \end{bmatrix} \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 1 \end{bmatrix}^{-1} \\
\left(\begin{bmatrix} 4 & -1 \\ 1 & 0 \end{bmatrix} \right)^n &= \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \phi_1^n & \\ & \phi_2^n \end{bmatrix} \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 1 \end{bmatrix}^{-1} \\
&\left(\text{where } \begin{cases} \phi_1 = 2 + \sqrt{3} \\ \phi_2 = 2 - \sqrt{3} \end{cases} \text{ are the solutions to the characteristic equation } \phi^2 = 4\phi - 1 \right)
\end{aligned}$$

于是我们有:

$$\begin{aligned}
\begin{bmatrix} D_n \\ D_{n-1} \end{bmatrix} &= \begin{bmatrix} 4 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} D_{n-1} \\ D_{n-2} \end{bmatrix} \\
&= \dots \\
&= \left(\begin{bmatrix} 4 & -1 \\ 1 & 0 \end{bmatrix} \right)^n \begin{bmatrix} D_0 \\ D_{-1} \end{bmatrix} \\
&= \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \phi_1^n & \\ & \phi_2^n \end{bmatrix} \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \phi_1^n & \\ & \phi_2^n \end{bmatrix} \left(\frac{1}{\phi_1 - \phi_2} \begin{bmatrix} 1 & -\phi_2 \\ -1 & \phi_1 \end{bmatrix} \right) \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} \phi_1^{n+1} & \phi_2^{n+1} \\ \phi_1^n & \phi_2^n \end{bmatrix} \left(\frac{1}{\phi_1 - \phi_2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) \\
&= \frac{1}{\phi_1 - \phi_2} \begin{bmatrix} \phi_1^{n+1} - \phi_2^{n+1} \\ \phi_1^n - \phi_2^n \end{bmatrix}
\end{aligned}$$

因此 D_n 的通项公式为:

$$D_n = \frac{\phi_1^{n+1} - \phi_2^{n+1}}{\phi_1 - \phi_2}$$

(where $\begin{cases} \phi_1 = 2 + \sqrt{3} \\ \phi_2 = 2 - \sqrt{3} \end{cases}$ are the solutions to the characteristic equation $\phi^2 = 4\phi - 1$)

显然 D_n 恒为正数, 且关于 n 严格单调递增.

根据递推公式 $D_n = 4D_{n-1} - D_{n-2}$ 可知 $D_n > 3D_{n-1}$ 对任意正整数 n 都成立.

(2) 逆矩阵

考虑 n 阶对称三对角阵 A :

$$A = \begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & & & \\ & & \ddots & \ddots & \\ & & & \ddots & 4 & 1 \\ & & & & 1 & 4 \end{bmatrix}_{n \times n}$$

$$\det(A) = D_n = \frac{\phi_1^{n+1} - \phi_2^{n+1}}{\phi_1 - \phi_2} \text{ where } \phi_{1,2} = 2 \pm \sqrt{3}$$

其逆矩阵为:

$$\begin{aligned}
A^{-1} &= \frac{1}{\det(A)} \text{adj}(A) \\
&= \frac{1}{\det(A)} \{ [(-1)^{i+j} \det(A[\{i\}^c, \{j\}^c])]\}_{i,j=1}^n \}^T
\end{aligned}$$

可以验证 A^{-1} 的第一列是:

$$(A^{-1})_{[:,1]} = \begin{bmatrix} D_{n-1} \\ -D_{n-2} \\ \vdots \\ (-1)^n D_1 \\ (-1)^{n+1} D_0 \end{bmatrix} = \frac{1}{\phi_1 - \phi_2} \begin{bmatrix} \phi_1^n - \phi_2^n \\ -(\phi_1^{n-1} - \phi_2^{n-1}) \\ \vdots \\ (-1)^n (\phi_1^2 - \phi_2^2) \\ (-1)^{n+1} (\phi_1 - \phi_2) \end{bmatrix}$$

- 邵老师: 可以根据

(3) 错误的 k_0 带来的误差

现考虑稀疏线性方程组:

$$\begin{bmatrix} 4\beta & \beta & & & \\ \beta & 4\beta & \beta & & \\ & \beta & \ddots & \ddots & \\ & & \ddots & 4\beta & \beta \\ & & & \beta & 4\beta \end{bmatrix}_{(n-1)} \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_{n-2} \\ k_{n-1} \end{bmatrix} = \begin{bmatrix} \eta_0 + \eta_1 - \beta k_0 \\ \eta_1 + \eta_2 \\ \vdots \\ \eta_{n-3} + \eta_{n-2} \\ \eta_{n-2} + \eta_{n-1} - \beta k_n \end{bmatrix}$$

$$\Updownarrow$$

$$\begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & 1 & \ddots & \ddots & \\ & & \ddots & 4 & 1 \\ & & & 1 & 4 \end{bmatrix}_{(n-1)} \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_{n-2} \\ k_{n-1} \end{bmatrix} = \frac{1}{\beta} \begin{bmatrix} \eta_0 + \eta_1 - \beta k_0 \\ \eta_1 + \eta_2 \\ \vdots \\ \eta_{n-3} + \eta_{n-2} \\ \eta_{n-2} + \eta_{n-1} - \beta k_n \end{bmatrix}$$

假设 k_0 带上误差 Δk_0 , 则在不考虑舍入误差的情况下我们有:

$$\begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & 1 & \ddots & \ddots & \\ & & \ddots & 4 & 1 \\ & & & 1 & 4 \end{bmatrix}_{(n-1)} \begin{bmatrix} k_1 + \Delta k_1 \\ k_2 + \Delta k_2 \\ \vdots \\ k_{n-2} + \Delta k_{n-2} \\ k_{n-1} + \Delta k_{n-1} \end{bmatrix} = \frac{1}{\beta} \begin{bmatrix} \eta_0 + \eta_1 - \beta(k_0 + \Delta k_0) \\ \eta_1 + \eta_2 \\ \vdots \\ \eta_{n-3} + \eta_{n-2} \\ \eta_{n-2} + \eta_{n-1} - \beta k_n \end{bmatrix}$$

于是我们有:

$$\begin{aligned}
\text{Denote } A &= \begin{bmatrix} 4 & 1 & & & \\ 1 & 4 & 1 & & \\ & 1 & \ddots & \ddots & \\ & & \ddots & 4 & 1 \\ & & & 1 & 4 \end{bmatrix}_{(n-1) \times (n-1)} \\
\begin{bmatrix} \Delta k_1 \\ \Delta k_2 \\ \vdots \\ \Delta k_{n-2} \\ \Delta k_{n-1} \end{bmatrix} &= A^{-1} \cdot \frac{1}{\beta} \begin{bmatrix} -\beta \Delta k_0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} \\
&= -\Delta k_0 \cdot A_{[:,1]}^{-1} \\
&= -\Delta k_0 \begin{bmatrix} D_{n-2} \\ -D_{n-3} \\ \vdots \\ (-1)^{n-1} D_1 \\ (-1)^n D_0 \end{bmatrix} \\
&= -\Delta k_0 \cdot \frac{1}{\phi_1 - \phi_2} \begin{bmatrix} \phi_1^{n-1} - \phi_2^{n-1} \\ -(\phi_1^{n-2} - \phi_2^{n-2}) \\ \vdots \\ (-1)^{n-1}(\phi_1^2 - \phi_2^2) \\ (-1)^n(\phi_1 - \phi_2) \end{bmatrix} \\
&\quad (\text{where } \phi_{1,2} = 2 \pm \sqrt{3})
\end{aligned}$$

因此我们有:

$$\begin{aligned}
\Delta k_i &= (-1)^i D_{n-i-1} \Delta k_0 = (-1)^i \frac{\phi_1^{n-i} - \phi_2^{n-i}}{\phi_1 - \phi_2} \Delta k_0 \\
&\quad \text{where } i = 1, 2, \dots, n-1
\end{aligned}$$

根据 (1) 的结论可知 D_n 是关于 n 严格单调递增的正数,

且 $D_n > 3D_{n-1}$ 对任意正整数 n 都成立.

因此由 Δk_0 导致的误差 $\Delta k_1, \Delta k_2, \dots, \Delta k_{n-2}, \Delta k_{n-1}$ 至少以 $\frac{1}{3}$ 的比例逐级递减.

Problem 7 (optional)

1

1. In the context of data fitting, a few nonlinear models are called intrinsically linear because they can be converted to linear models by certain transformations. Show that the following nonlinear models are intrinsically linear (under mild assumptions) by finding appropriate transformations.

(1) $y = a \exp(bx + cx^2)$.

(2) $y = 1/(1 + \exp(a + bx))$.

(3) $y = ax/(b + \sqrt{x})^2$.

解: 下面都假设有 n 个数据点 $(x_1, y_1) \dots (x_n, y_n)$

(1) $y = a \exp(bx + cx^2)$

$\Rightarrow \ln y = \ln a + bx + cx^2$

令 $\tilde{a} = \ln a$, 则:

$\ln y = \tilde{a} + bx + cx^2$

解以下最小二乘问题:

$$\begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix} \begin{bmatrix} \tilde{a} \\ b \\ c \end{bmatrix} = \begin{bmatrix} \ln y_1 \\ \ln y_2 \\ \vdots \\ \ln y_n \end{bmatrix}$$

得到最小二乘解 $\tilde{a}, b, c$ 后令 $a = e^{\tilde{a}}$ 即可

(2) $y = 1/(1 + \exp(a + bx))$

$\Rightarrow \frac{1}{y} = 1 + \exp(a + bx)$

$\ln\left(\frac{1}{y} - 1\right) = a + bx$

则 $\ln(\bar{y}-1)$ 关于 x 是线性的

$$(3) \quad y = ax / (b + \sqrt{x})^2$$

$$\Rightarrow \frac{ax}{y} = (b + \sqrt{x})^2$$

$$\Rightarrow \sqrt{a} \cdot \sqrt{\frac{x}{y}} = b + \sqrt{x}$$

$$\Rightarrow \sqrt{\frac{x}{y}} = \frac{b}{\sqrt{a}} + \frac{\sqrt{x}}{\sqrt{a}}$$

则 $\sqrt{\frac{x}{y}}$ 关于 $\sqrt{x}$ 是线性的

□

假设 $x, y, a > 0$:

为了将非线性方程 $y = \frac{ax}{(b+\sqrt{x})^2}$ 转换为线性关系，可以通过变量替换实现。以下是关键步骤：

转换方法

1. 变量替换：

令 $t = \sqrt{x}$ ，则 $x = t^2$ 。原方程变为：

$$y = \frac{at^2}{(b+t)^2}.$$

2. 取倒数并开平方：

对两边取倒数后开平方：

$$\sqrt{\frac{1}{y}} = \frac{b+t}{\sqrt{a}t} = \frac{1}{\sqrt{a}} + \frac{b}{\sqrt{a}t}.$$

3. 定义新变量：

设 $u = \sqrt{\frac{1}{y}}$ 和 $v = \frac{1}{t} = \frac{1}{\sqrt{x}}$ ，则方程可改写为：

$$u = \frac{1}{\sqrt{a}} + \frac{b}{\sqrt{a}}v.$$

4. 线性关系：

此时， u 与 v 之间为线性关系：

$$u = mv + c,$$

其中斜率 $m = \frac{b}{\sqrt{a}}$ ，截距 $c = \frac{1}{\sqrt{a}}$ 。

参数求解

通过线性回归确定 m 和 c 后，原参数 a 和 b 可表示为：

$$a = \frac{1}{c^2}, \quad b = \frac{m}{c}.$$

结论

在变量替换 $u = \sqrt{\frac{1}{y}}$ 和 $v = \frac{1}{\sqrt{x}}$ 下，原方程转换为线性关系：

$$u = \frac{1}{\sqrt{a}} + \frac{b}{\sqrt{a}}v.$$

这种形式允许使用线性回归法估计参数 a 和 b 。

Problem 8 (optional)

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2. Chebyshev polynomials can be expressed as $T_n(x) = \cos(n \arccos x)$ for $x \in [-1, 1]$. Can you find an explicit expression of $T_n(x)$ for $x \in \mathbb{R} \setminus [-1, 1]$ in terms of algebraic functions?

解: Chebyshev 多项式有如下递推关系:

$$T_n = 2x T_{n-1} - T_{n-2} \quad x \in \mathbb{R} \setminus [-1, 1]$$

解特征方程: $\lambda^2 = 2x\lambda - 1$

$$\Rightarrow \lambda_1 = \frac{2x + \sqrt{4x^2 - 4}}{2} = x + \sqrt{x^2 - 1}$$

$$\lambda_2 = \frac{2x - \sqrt{4x^2 - 4}}{2} = x - \sqrt{x^2 - 1}$$

$$\text{设 } T_n = \mu_1 \cdot \lambda_1^n + \mu_2 \cdot \lambda_2^n$$

由:

$$\begin{cases} T_0 = \mu_1 + \mu_2 = 1 \\ T_1 = \mu_1 \lambda_1 + \mu_2 \lambda_2 = x \end{cases}$$

$$\Rightarrow \mu_1 = \mu_2 = \frac{1}{2}$$

$$\Rightarrow T_n = \frac{1}{2} (x + \sqrt{x^2 - 1})^n + \frac{1}{2} (x - \sqrt{x^2 - 1})^n$$

$$x \in \mathbb{R} \setminus [-1, 1]$$

□